

Journal Supplementary D — Full mathematical derivations

Companion to: *Beyond Accuracy* (Section IV)

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A.1 Scope

This supplement provides the complete mathematical content for the twelve-axis framework: generative models, variational inference, hierarchical Bayesian likelihood, coherence formulae, ARI and Hungarian definitions, and the algorithmic pseudocode for each axis.

B.2 Canonical generative model

B.2.1 LDA on band-frequency documents

For K topics, vocabulary $V = B$ band indices, D documents and hyperparameters $\alpha, \eta > 0$:

$$\phi_k \sim \text{Dir}(\eta \mathbf{1}_V), \quad k = 1, \dots, K, \quad (1)$$

$$\theta_d \sim \text{Dir}(\alpha \mathbf{1}_K), \quad d = 1, \dots, D, \quad (2)$$

$$z_{d,n} \sim \text{Cat}(\theta_d), \quad n = 1, \dots, N_d, \quad (3)$$

$$w_{d,n} \sim \text{Cat}(\phi_{z_{d,n}}). \quad (4)$$

B.2.2 Online VB lower bound

The mean-field variational family is $q(\theta_d, z_d, \phi) = q(\theta_d \mid \gamma_d)(\prod_n q(z_{d,n} \mid \lambda_{d,n})) \prod_k q(\phi_k \mid \beta_k)$. The per-document ELBO is

$$\begin{aligned} \mathcal{L}_d = & \sum_n \sum_k \lambda_{d,n,k} [\psi(\gamma_{d,k}) - \psi(\sum_{k'} \gamma_{d,k'}) \\ & + \psi(\beta_{k,w_{d,n}}) - \psi(\sum_v \beta_{k,v}) - \log \lambda_{d,n,k}] \\ & + (\alpha - 1) \sum_k [\psi(\gamma_{d,k}) - \psi(\sum_{k'} \gamma_{d,k'})] \\ & + \log \Gamma(\sum_{k'} \gamma_{d,k'}) - \sum_k \log \Gamma(\gamma_{d,k}) + \text{const.} \end{aligned} \quad (5)$$

Updates are Eq. (8) and Eq. (9) of the conference supplement B. Online VB applies the natural-gradient ascent $\beta \leftarrow (1 - \rho_t)\beta + \rho_t\tilde{\beta}_t$ with $\rho_t = (\tau_0 + t)^{-\kappa}$, $\tau_0 = 10$, $\kappa = 0.7$.

B.2.3 Posterior summaries

$$\phi_{k,b} = \beta_{k,b} / \sum_v \beta_{k,v}; \theta_{d,k} = \gamma_{d,k} / \sum_{k'} \gamma_{d,k'}; z_d = \arg \max_k \theta_{d,k}.$$

C.3 Axis F-1: hierarchical Bayesian model

C.3.1 Likelihood

For M methods, S scenes, F folds, observed fold-OA scores $y_{m,s,f}$:

$$y_{m,s,f} = \mu_m + \delta_s + \rho_f + \epsilon_{m,s,f}, \quad (6)$$

$$\mu_m \sim \mathcal{N}(0, 1), \quad (7)$$

$$\delta_s \sim \mathcal{N}(0, 0.5^2), \quad (8)$$

$$\rho_f \sim \mathcal{N}(0, 0.2^2), \quad (9)$$

$$\epsilon_{m,s,f} \sim \mathcal{N}(0, \sigma^2), \quad (10)$$

$$\sigma \sim \text{HalfNormal}(0.5). \quad (11)$$

C.3.2 Inference

NUTS with 1000 draws + 1000 tune, 2 chains, $\hat{R} < 1.01$ on all parameters in all reported runs. Posterior mean and 94% HDI of μ_m are reported; pairwise $P[\mu_A > \mu_B]$ are estimated from the joint posterior draws as

$$P[\mu_A > \mu_B] \approx \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\mu_A^{(t)} > \mu_B^{(t)}], \quad (12)$$

with $T = 2000$ draws across two chains.

D.4 Axis F-2: coherence measures

D.4.1 c_v coherence

For a topic t with top- N words $W_t = (w_1, \dots, w_N)$, the NPMI between words w_i, w_j on a sliding window of size s is

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{P(w_i, w_j) + \epsilon}{P(w_i)P(w_j)}}{-\log(P(w_i, w_j) + \epsilon)}. \quad (13)$$

The topic context vector is $\mathbf{v}_t = \sum_{i=1}^N \mathbf{u}_i$ with $u_{i,j} = \text{NPMI}(w_i, w_j)$. The c_v score is

$$c_v(t) = \frac{1}{N} \sum_{i=1}^N \cos(\mathbf{v}_t, \mathbf{u}_i), \quad (14)$$

the average cosine similarity between the topic context vector and each word's individual context vector. We use $N = 15$ and the default Röder sliding window of size $s = 110$.

D.4.2 NPMI and U-Mass

$$c_{\text{NPMI}}(t) = \binom{N}{2}^{-1} \sum_{i < j} \text{NPMI}(w_i, w_j). \quad \text{U-Mass}(t) = \binom{N}{2}^{-1} \sum_{i < j} \log \frac{N(w_i, w_j) + 1}{N(w_j)}.$$

E.5 Axis F-3: seed stability

For each (scene, method): refit five LDA / ProdLDA / ETM under seeds $\{0, \dots, 4\}$. For each refit compute KMeans clustering of θ into K clusters; report $\text{ARI}(z_{\text{KMeans}}, \text{label})$, c_v . Aggregate across seeds as mean $\pm$ std.

F.6 Axis F-4: capacity sensitivity

Refit LDA at $K \in \{4, 6, 8, 10, 12, 16, 20, 24\}$ with the canonical hyperparameters, $N = 5$ seeds per K . Per- K aggregates:

- perplexity $_K^{\text{test}} = \exp(-\frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{d \in \mathcal{D}_{\text{test}}} \log p(\mathbf{w}_d | \phi_K, \alpha))$,
- topic-diversity $_K = \frac{|\bigcup_k W_k^{(K)}|}{N \cdot K}$ with $W_k^{(K)}$ the top- N words per topic at capacity K ,
- matched-cosine $_K = \text{mean over (seed pair, topic) of } \cos(\phi_k^{(s_1)}, \phi_{\sigma^*(k)}^{(s_2)})$.

G.7 Axis F-5: paired ARI under masks

Detailed in conference supplement B; restated here for cross-reference. $\text{ARI}_{\text{paired}}(m) = \text{ARI}(\{z_d\}_S, \{z_d^{(m)}\}_S)$.

H.8 Axis F-6: cross-method ARI

For each scene compute the partition set $\mathcal{P} = \{\text{label, topic-dom, Felz, SLIC-500, SLIC-2000, p-15, p-7, px}\}$. The reported matrix is

$$M_{ij} = \text{ARI}(\mathcal{P}_i, \mathcal{P}_j), \quad (15)$$

restricted to pixels with a valid (non-sentinel) value in both partitions. NMI matrix is computed analogously.

I.9 Axis F-7: topic-label coupling

For each (canonical topic t , label ℓ), $P(\ell | t) = N_{t,\ell} / \sum_{\ell'} N_{t,\ell'}$ with $N_{t,\ell}$ the count of pixels with dominant topic t and ground truth ℓ . Per-topic Shannon entropy: $H(L | t) = -\sum_{\ell} P(\ell | t) \log P(\ell | t)$. Per-topic KL divergence from the marginal: $\text{KL}(P(L | t) \| P(L)) = \sum_{\ell} P(\ell | t) \log[P(\ell | t) / P(\ell)]$.

J.10 Axis F-8: per-topic Hungarian alignment

Confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$, $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$. Hungarian: $\sigma^* = \arg \max_{\sigma} \sum_k C_{k,\sigma(k)}$, solved by `scipy.optimize.linear_sum_assignment` on $-C$. Swap rate: $\text{swap} = |\mathcal{S}|^{-1} \sum_d \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$.

K.11 Axis F-9: HIDSAG preprocessing stability

For each HIDSAG subset, $R = 3$ preprocessing recipes $\{\text{cont-rem, deriv, smooth}\}$. For each recipe pair (r_a, r_b) compute $\text{ARI}(\text{dom-top}_{r_a}, \text{dom-top}_{r_b})$ on the documents present in both recipes. Per-subset aggregate is the mean over the $\binom{3}{2} = 3$ pairs.

L.12 Axis F-10: cross-scene transfer

Given scene s with canonical basis $\phi^{(s)}$, project the documents of scene t through one E-step

$$\gamma_{d,k}^{(t \leftarrow s)} = \alpha + \sum_n \lambda_{d,n,k}^{(t \leftarrow s)}, \quad (16)$$

with λ computed from $\phi^{(s)}$ frozen. The transfer report is $\text{ARI}(\text{label}_t, \arg \max_k \theta^{(t \leftarrow s)})$.

M.13 Axis F-11: rate–distortion

Reconstruct $\hat{\mathbf{x}}_d = \sum_k \theta_{d,k} \phi_k^\top$ in the band-frequency-count space, then de-quantise to a continuous reflectance via the inverse of the canonical tokenisation $\hat{x}_{d,b} = c_{d,b}^{\text{recon}}/Q$. Distortion is $\text{MSE} = D^{-1} \sum_d \|\hat{\mathbf{x}}_d - \mathbf{x}_d\|^2$ at rate $\log_2 K$ bits.

N.14 Axis F-12: external baseline

For each labelled scene, collect three reference OA values from prior published work. Report $\Delta_s = \text{OA}_s^{\text{topic-routed-soft}} - \max_{r \in \text{refs}(s)} \text{OA}_s^{(r)}$ with the per-paper split caveat.

O.15 Inverse Hungarian (used in F-8 swap-rate computation)

The post-permutation confusion matrix $\tilde{C}$ used for the journal Figure 3(b) panel is

$$\tilde{C}_{k,j} = C_{k,\sigma^{*-1}(j)}, \quad (17)$$

which moves the matched diagonal to the main diagonal. The right panel of journal Figure 3 visualises $\tilde{C}$ directly.

References